Unit 3 - Session 6

SLO 1: Generalized Hough Transform (GHT)- Model arbitrary shape detection using reference points

Duration: 10 Minutes Activity

Question:

Why Classical Hough Fails Here?"

Answer: Classical Hough Transform is limited because it requires a predefined mathematical equation for the shape. It can detect circles and lines/rectangles, but it cannot easily detect an irregular logo shape without such an equation.

Step 1

Show 3 objects:

1. A circle
2. A rectangle
3. An irregular logo shape

Ask:

Can classical Hough detect this using a simple equation?

Students answer YES for:

- Circle
- Line/rectangle (using line equations)

Students struggle with:

- Irregular logo

Step 2 (4 min)

Ask:

If there is no equation, how can we detect it?

Answer: Store the boundary points, store their relative positions, and use a reference point. The relative vector to the center forms the basis of the R-table idea.

Force them to propose:

- Store boundary points
- Store relative positions
- Use reference point

Guide them toward:
→ "Store relative vector to center"
→ That becomes the R-table idea

Step 3

Ask:

- What is the major limitation of classical Hough?

Answer: Its major limitation is that it is mainly designed for shapes that can be represented by known mathematical equations.

- Why is GHT more general?

Answer: GHT is more general because it can detect arbitrary shapes by storing boundary-point information and relative vectors in an R-table instead of requiring a fixed mathematical equation.

- What is the cost of this flexibility?

Answer: The flexibility of GHT comes at the cost of increased computation and memory because it stores and searches more shape information.